

Algorithms 2

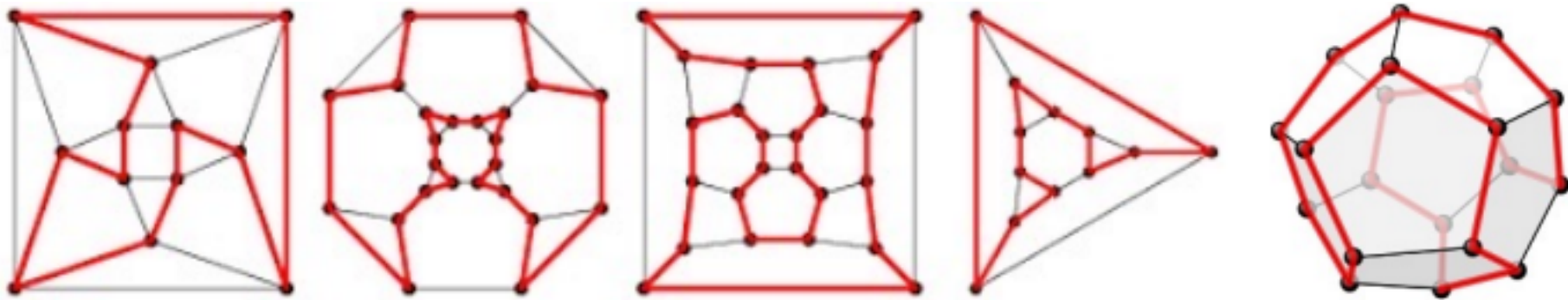
Hamiltonicity

Daniel Rosenberg & Michael Trushkin

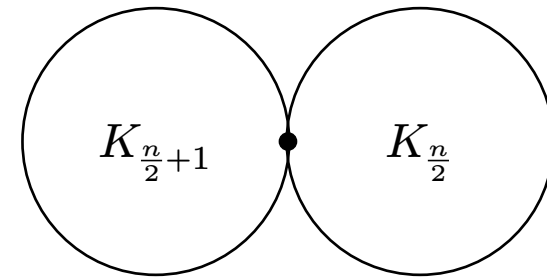
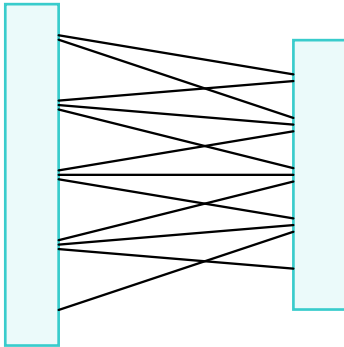
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1. Dirac's theorem

- A simple path spanning $V(G)$ is called a *Hamilton* path
- A simple cycle spanning $V(G)$ is called a *Hamilton* cycle
- A graph containing a Hamilton path is called *traceable*
- A graph containing a Hamilton cycle is called *Hamiltonian*



Here are two examples for **non**-hamiltonian graph



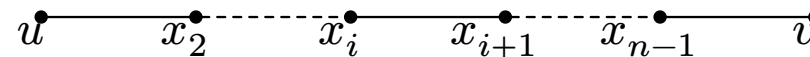
Question 1.1.1 What is common among the two examples?

Answer: They are both quite dense: $\delta(G) = \frac{n}{2} - 1$

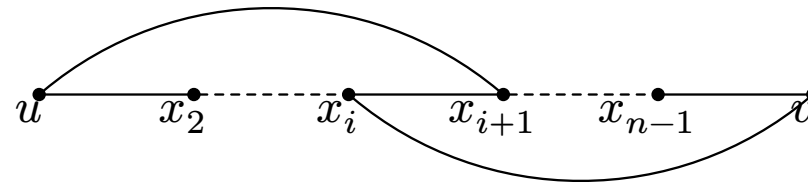
Theorem 1.1.2 (Dirac's theorem) Let G have $v(G) \geq 3$ and $\delta(G) \geq \frac{n}{2}$. then G is Hamiltonian

proof:

- Suppose the claim is false
- let G be an edge maximal counter example:
 - G does not contain an Hamiltonian cycle
 - $G + e$ is Hamiltonian $\forall e \notin E(G)$
- Let $e = uv \notin E(G)$
- As $G + e$ is Hamiltonian, G contains a Hamilton path



- If we could find an edge $x_i x_{i+1}$ such that
 - $ux_{i+1} \in E(G)$
 - $vx_i \in E(G)$
- Then we can reroute our path, and turn it into Hamilton cycle



- Define $S := \{i \in [n] : ux_{i+1} \in E(G)\}$
 - note that $n \notin S$
 - but $1 \in S$ as $ux_1 \in E(G)$
- Define $T := \{i \in [n] : vx_i \in E(G)\}$
 - note the $n \notin S$ as G is simple

-If $|S \cap T| \neq \emptyset$ we can reroute

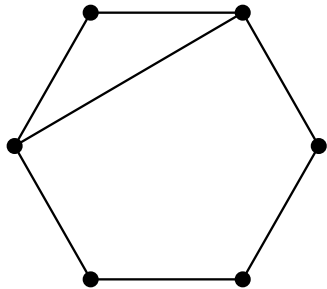
- Define $S := \{i \in [n] : ux_{i+1} \in E(G)\}$
 - note that $n \notin S$
 - but $1 \in S$ as $ux_1 \in E(G)$
- Define $T := \{i \in [n] : vx_i \in E(G)\}$
 - note the $n \notin S$ as G is simple
- Assume the $|S \cap T| = 0$

$$|S \cup T| = |S| + |T| - \underbrace{|S \cap T|}_0 \geq \deg(u) + \deg(v) \geq \frac{n}{2} + \frac{n}{2} = n$$

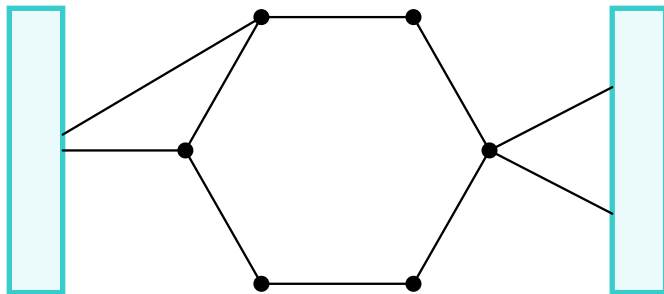
- We got $|S \cup T| \geq n$
- As $n \notin S \cup T$, we get a contradiction.

2. Eros-Chvatal

- Let C be a cycle in G .
- By a bridge of C we mean:
 - an edge between to vertices of C , called a cord, those are trivial bridges

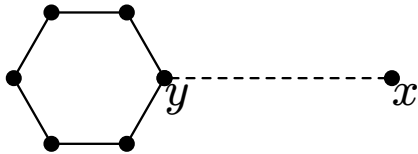


- connected components of $G - C$ that has neighbors in C , those are non-trivial bridges



Definition 2.1.1

- Let P be a $x \rightsquigarrow y$ path in G
- Let C be a cycle in G
- If $y \in C$ then $C \cup P$ is called (x, y) -lollipop

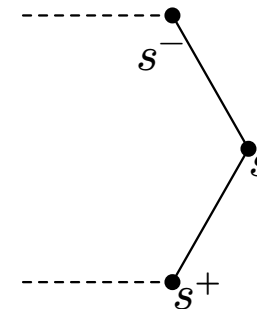


- Let $v \in V(C)$ and orient C
 - Denote by v^+ the successor of v
 - Denote by v^- the predecessor of v

For $S \subseteq V(C)$ write

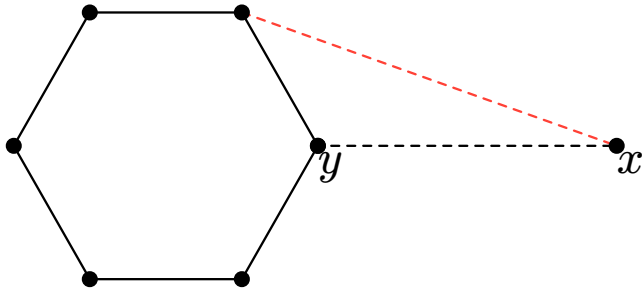
$$S^+ := \{s^+ : s \in S\}$$

$$S^- := \{s^- : s \in S\}$$



Lemma 2.1.2

- Let C be the largest cycle in G
- Let $C \cup P$ be an (x, y) -lollipop
- The edges xy^+ and xy^- do not exist in G



Theorem 2.2.1 (Erdos Chvatal Theorem)

- Let G have $v(G) \geq 3$
- If $\kappa(G) \geq \alpha(G)$ then G is Hamiltonian.

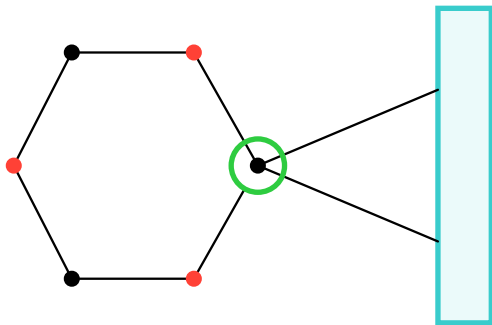
proof:

- Assume the claim is false
- Let C be the longest cycle in G
- Why such cycle exists?
 - If G is disconnected the $0 = \kappa(G) \geq \alpha(G) = 2$, contradiction
 - If G is a connected tree then $1 = \kappa(G) = \alpha(G) = 2$
- As G is not hamiltonial $V(G) \setminus V(C) \neq \emptyset$
- Then G admits a non-trivial bridge with ≥ 2 points of attachment
- Why?

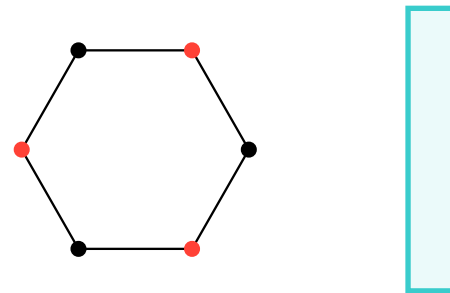
Theorem 2.2.2 (Erdos Chvatal Theorem)

- Let G have $v(G) \geq 3$
- If $\kappa(G) \geq \alpha(G)$ then G is Hamiltonian.

- Then G admits a non-trivial bridge with ≥ 2 points of attachment
- Why?
 - Assume the bridge has < 2 points of attachments
 - Observe the components of $G - C$, one of the two can happen



$1 = \kappa(G) \geq \alpha(G) \geq 2$, Contridiction

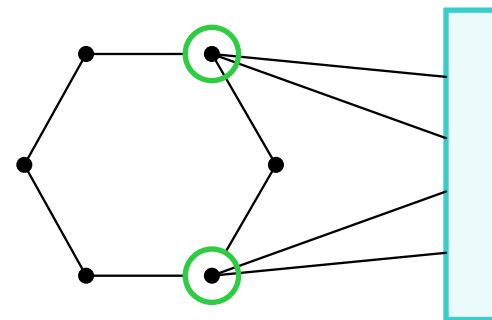


$0 = \kappa(G) \geq \alpha(G) \geq 2$, Contridiction

Theorem 2.2.3 (Erdos Chvatal Theorem)

- Let G have $v(G) \geq 3$
- If $\kappa(G) \geq \alpha(G)$ then G is Hamiltonian.

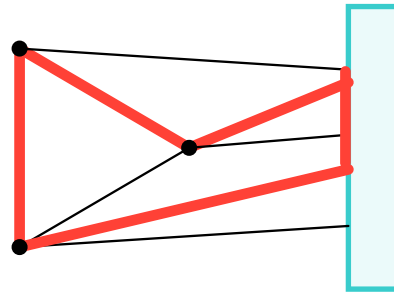
- Let B a bridge as before
- Let S be the points of attachment of B
- Notice the every $b \in B$ and $s \in S$ form a (b, s) -lollipop
- Orient C
- Notice that S^+ is independent
 - If this is not the case, we can increase C
- By the lollipop lemma: $\forall b \in B, s \in S : bs^+ \notin E(G)$
- Hence $s^+ \cup \{b\}$ is independent for $b \in B$
- $\alpha(G) \geq |S| + 1$



Theorem 2.2.4 (Erdos Chvatal Theorem)

- Let G have $v(G) \geq 3$
- If $\kappa(G) \geq \alpha(G)$ then G is Hamiltonian.

- If S does not form a vx-cut then $V(C) = S$



- As the set S is a vx-cut in G :

$$\kappa(G) \leq |S| \leq |S^+| < |S^+| + 1 = \alpha(G)$$

contradiction.